



## Handout 6.3: Waste Audit Data Sheet and Scale Up

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

Print pages 1 to 3 for every student. Page 4 is the sample data set for classes that could not collect their own, and for grade 6. Use the **grams** column if you weighed, and the **items** column if you counted. Every calculation works the same way in either unit.

## Part 1. Where food waste happens

| Where                         | One real example | About what share of all food waste |
|-------------------------------|------------------|------------------------------------|
| On the farm                   |                  |                                    |
| At the store                  |                  |                                    |
| At home                       |                  |                                    |
| In cafeterias and restaurants |                  |                                    |

Circle the one that surprised you.

My guess before the lesson: \_\_\_\_\_ pounds per person per year. The real number is about: \_\_\_\_\_ pounds per person per year.

## Part 2. Our audit

**Our method** (circle one): we weighed / we counted items

**Days collected:** \_\_\_\_\_ **Number of people the bins came from:** \_\_\_\_\_

| Bin                                       | Weight in grams | Weight in ounces | Number of items |
|-------------------------------------------|-----------------|------------------|-----------------|
| EDIBLE (somebody could have eaten it)     |                 |                  |                 |
| NOT EDIBLE (peels, cores, bones, grounds) |                 |                  |                 |
| <b>TOTAL</b>                              |                 |                  |                 |

**The edible bin is the one we do the math on.** Peels and cores were never going to be eaten, so they are not the number we are trying to change.

The one thing in the edible bin that bothered me most: \_\_\_\_\_

### Part 3. The math

Show your work in every box. Label every answer with its unit.

#### Calculation 1: per person per day

Total edible waste \_\_\_\_\_ divided by number of people \_\_\_\_\_ = \_\_\_\_\_ per person for all the days

Then divide by the number of days \_\_\_\_\_ = \_\_\_\_\_ **per person per day**

Work:





### Calculation 2: per person per school year

Per person per day \_\_\_\_\_ times 180 school days = \_\_\_\_\_ **per person per school year**

Work:



Convert to pounds: divide grams by 454, or divide ounces by 16.

\_\_\_\_\_ **pounds per person per school year**

**Calculation 3: the whole school**

Per person per school year \_\_\_\_\_ times number of students in the building \_\_\_\_\_  
= \_\_\_\_\_

In pounds: \_\_\_\_\_ **pounds of edible food per school year, whole school**

Work:





**Check your units.** Circle the unit of your final answer: grams / ounces / pounds / items

## Part 4. Three changes

Each change has to name **who** does **what** and **when**. “Waste less” is not a change.

| # | Who | What they do | When |
|---|-----|--------------|------|
| 1 |     |              |      |
| 2 |     |              |      |
| 3 |     |              |      |

Which of your three costs nothing? \_\_\_\_\_

Which one would still be happening in a month? \_\_\_\_\_ Why that one?



**Class vote result, the three changes we are actually doing:**







### Exit card

Our class wastes about \_\_\_\_\_ pounds a year, and the one change I will actually do is \_\_\_\_\_.

## Page 4: Sample data set

Use this if your class could not collect its own data. Say out loud that it is sample data and not ours.

A class of 24 students collected food scraps for 2 days. EDIBLE bin: 1,440 grams total. NOT EDIBLE bin: 2,160 grams total. The building has 600 students.

Work it through:

1,440 divided by 24 = \_\_\_\_ grams per person for 2 days \_\_\_\_ divided by 2 = \_\_\_\_ grams per person per day \_\_\_\_ times 180 = \_\_\_\_ grams per person per school year \_\_\_\_ divided by 454 = about \_\_\_\_ pounds per person per school year \_\_\_\_ times 600 students = about \_\_\_\_ pounds for the whole school

**Grade 6 version:** do the first four lines only. Stop before the whole-school line.

## Page 5: Grade 8 stretch, put a dollar on it

Use an average food price per pound from this week's store circular.

Average price per pound I am using: \$\_\_\_\_\_ (what food? \_\_\_\_\_)

| Question                                               | Work | Answer |
|--------------------------------------------------------|------|--------|
| What does one person's yearly edible waste cost?       |      | \$     |
| What does the whole school's yearly edible waste cost? |      | \$     |
| What could the FACS room buy with that?                |      |        |

**One sentence:** the number that surprised me most was \_\_\_\_\_ because \_\_\_\_\_.

## Teacher key

### Part 1 answers (use current figures the week you teach it):

| Where                         | Example                                                                                                                                       | Share                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| On the farm                   | A crop left in the field because the price dropped, a harvest lost to weather, or produce rejected for size or shape                          | roughly 15 to 20 percent of food loss |
| At the store                  | Produce pulled for looks, bakery and deli items pulled at closing, food pulled at a best-by date that is a quality date and not a safety date | roughly 10 percent                    |
| At home                       | Food bought and forgotten, leftovers not eaten, portions too large. This is the single biggest piece in the United States                     | roughly 40 to 50 percent              |
| In cafeterias and restaurants | Whole unopened items discarded, plate waste, over-prepared batches                                                                            | roughly 20 to 25 percent              |

Overall: about a third of the food produced in this country is never eaten, and the per-person figure most commonly cited is roughly 200 to 300 pounds a year of food thrown away.

**Part 3, sample data answers:** 60 grams per person for 2 days; 30 grams per person per day; 5,400 grams per person per school year; about 12 pounds per person per school year; about 7,200 pounds, or about 3.6 tons, for a 600 student school.

### The two errors to watch for:

1. Multiplying by 180 before dividing by the number of days. The redirect is one question: “is 60 grams one day or two?”
2. Losing the unit. Ask “grams or ounces?” at every table. A number without a unit is not an answer.

### Part 4, what a real change looks like:

- “The Station Manager puts unopened leftovers in the labeled bin at the end of every lab, and the next class gets first call on them.”
- “Toppings Chief measures cheese into a cup instead of pouring from the bag, starting with the next lab.”

- “Whoever sets the trash bin puts the compost bin next to it, so peels do not go in the trash by default.”
- “At lunch, anyone who does not want their unopened milk puts it on the share table instead of the tray.”

Not a change: “everybody should be less wasteful,” “the cafeteria should make better food,” “recycle more.”

**Grade 8 stretch, sample:** at \$2.00 a pound, 12 pounds per person is about \$24 a year per student, and 7,200 pounds is about \$14,400 for a 600 student school. That number is the one that lands.

**Scoring:** the data sheet is daily work, scored complete, partial, or missing. The two things to look for: correct units on all three calculations, and three changes that each name a who, a what, and a when.

**One handling note:** the class number is a class number. Do not report per-station or per-student waste in a way that identifies who wasted what. If one lab’s scraps are much larger, say it as a recipe problem, not a behavior problem.